

An Adaptive QoS-Aware Priority Scheduling Solution for Dynamic Radio Resource Allocation in Multiservice 5G Network Slicing Environments

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Abstract—Fifth generation (5G) radio access network (RAN) slicing provides support for resource isolation and dynamic resource optimization to accommodate the diverse quality of service (QoS) needs across Internet of Things (IoT) ecosystems, i.e., enhanced mobile broadband (eMBB), ultrareliable low-latency communications (URLLC), and massive machine type communication (mMTC) or best effort (BE) services. However, existing optimization- and data-driven radio resource allocation (RRA) models often suffer from rigid/suboptimal resource distribution and increased service level agreement (SLA) violations in dynamic, multiservice environments. This article introduces an adaptive QoS-aware prioritized resource scheduling (AQPS) solution, a novel RRA approach designed to optimize resource distribution for multiservice 5G network slicing deployments in IoT-enabled infrastructures. To optimally allocate resource block groups (RBGs) according to time-varying traffic patterns and IoT network conditions, we formulate an NP-hard optimization problem with the objective of minimizing SLA violations while satisfying QoS requirements for diverse IoT service classes. In this context, AQPS optimizes resource distribution for IoT services by incorporating: 1) minimum guarantee allocation (QoS-driven RBG estimation); 2) weighted urgency-based resource distribution (computing user urgency by incorporating buffer state, QoS factors, spectral efficiency, and service priorities); and 3) priority-based round-robin (RR) allocation. Extensive trace-driven simulation findings from two comprehensive multiuser IoT-dense urban and enhanced IoT coverage zone scenarios with realistic channel conditions demonstrate that the AQPS solution experiences a very low number of SLA violations while enabling SLA assurance rates of 99.07% and 97.58%, respectively, when compared against state-of-the-art benchmarks, including deep reinforcement learning (DRL), round robin (RR), and stepwise optimal algorithm (SOA).

Index Terms—Fifth generation (5G), network slicing, quality of service (QoS) scheduling, radio access network (RAN) slicing, radio resource allocation (RRA), service level agreement (SLA) assurance.

NOMENCLATURE

3GPP	Third generation partnership project.
5G	Fifth generation.
IoT	Internet of Things.

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RAN	Radio access network.
O-RAN	Open radio access network.
UE	User equipment.
eMBB	Enhanced mobile broadband.
URLLC	Ultrareliable low-latency communications.
mMTC	Massive machine type communication.
BE	Best effort.
RRA	Radio resource allocation.
RRM	Radio resource manager.
RB	Resource block.
RBG	RB group.
TTI	Transmission time interval.
AQPS	Adaptive QoS-aware prioritized resource scheduling solution.
SOA	Stepwise optimal algorithm.
RR	Round robin.
RL	Reinforcement learning.
DRL	Deep RL.
DQN	Deep Q-network.
DDPG	Deep deterministic policy gradient.
RDGP	Recurrent deterministic policy gradient.
PPO	Proximal policy optimization.
SAC	Soft actor-critic.
CNN	Convolutional neural network.
QoS	Quality of service.
SLA	Service level agreement.
WPU	Worst-performing user.
BPU	Best-performing user.
QuaDRiGa	QUAsi Deterministic RadIo channel Gen-erator.

I. INTRODUCTION

IN RECENT years, RAN slicing has emerged as a vital technology for supporting numerous IoT services with diverse SLAs on shared infrastructure, necessitating adaptability, better resource management, and options for customization [1], [2], [3]. SLAs establish specific requirements that network operators must fulfill to deliver the appropriate QoS for users' applications [4]. In this context, advanced network slicing architectures offer an effective solution to meet the diverse SLA requirements of different network slices [5], [6]. For instance, eMBB slices require high throughput but can accommodate immediate latency. On the contrary, the ultrareliable low-latency communication (URLLC) slices demand strict

latency and reliability guarantees. At the same time, BE slices could have more relaxed requirements but still need minimum throughput assurance [7], [8].

Traditional network architectures, however, employ a standardized approach that restricts flexibility for implementing network slicing for various IoT services [9], [10], [11]. Emerging architectures like O-RAN address these limitations through disaggregation and standardized interfaces [12], [13]. However, sharing limited frequency spectrum resources among network slices with strict QoS demands in multiservice IoT deployments remains difficult as heterogeneous service demands continue to grow [14], [15]. This makes RBG's allocation challenging since each slice must get specific resources to meet its SLA while dealing with insufficient and limited frequency spectrum resources for IoT connectivity.

The diverse scenarios and requirements emerging in 5G and beyond networks, including dense urban IoT communication, high-demand industrial IoT automation, volumetric video streaming, high-data infotainment, healthcare IoT applications, autonomous vehicles, smart agriculture, etc., significantly increase the complexity of resource management [16], [17]. Therefore, efficient RRA in RAN slicing environments is crucial when balancing heterogeneous QoS requirements of different IoT service types. This is because achieving optimal resource utilization efficiency is complicated by fluctuating user associations with base stations due to high mobility patterns and highly dynamic wireless channel conditions [18], [19], [20]. In addition, adaptability to traffic patterns and network dynamics requires sophisticated state-of-the-art mechanisms that can respond in real-time to rapidly changing conditions without noticeable service degradation [21]. Therefore, maintaining strict SLA compliance while minimizing time-based violations becomes particularly demanding when distributing limited frequency spectrum resources among competing slices [22]. Current optimization- and data-driven RRA approaches [23], [24], [25] in network slicing often ignore SLA-specific QoS requirements and inefficiently allocate all resources regardless of demand while struggling with the operational complexity and coordination demands inherent in O-RAN architectures and learning-based implementations [26]. This multidimensional challenge necessitates cutting-edge resource distribution strategies that can smartly distribute resources while maintaining service guarantees across heterogeneous network deployments.

This article proposes an adaptive QoS-aware prioritized resource scheduling (AQPS) solution for dynamic and intelligent resource management in multiservice 5G network slicing environments. The proposed AQPS solution addresses critical RRA limitations by managing heterogeneous QoS requirements in IoT-centric deployments. Unlike existing static resource allocation and scenario-limited strategies [27], [28] or data-driven methods [15], [29], [30], [31], which require extensive hyperparameter tuning, reward engineering, and training periods under different SLA configurations, AQPS systematically computes immediate resource demands by employing throughput and buffer-driven latency constraints instead of minimum requirements [23] or greedy allocations [32]. Instead of solely relying on spectral efficiency [33], AQPS employs

a parameter-less, comprehensive, dynamic urgency-based prioritization mechanism that significantly improves resource utilization efficiency while maintaining strict SLA compliance in resource-constrained IoT environments. A multiphase allocation strategy of AQPS (described in Section IV-C) ensures best-fit resource distribution through guaranteed minimum connectivity, weighted urgency-based allocation to balance user-specific QoS requirements, and priority-based RR distribution in order to maximize resource utilization efficiency while respecting service hierarchies. The low-complex AQPS architecture dynamically adapts to changing conditions and enables efficient and scalable resource allocation without requiring extensive training or hyperparameter adjustments.

The main contributions of the proposed AQPS solution are as follows.

- 1) It introduces a novel QoS-driven resource demand estimation approach (Section IV-A) that converts heterogeneous QoS metrics (throughput and buffer urgency) into standardized resource requirements through spectral efficiency normalization for each scheduling interval.
- 2) It deploys a comprehensive urgency-based prioritization mechanism (Section IV-B) that computes user urgency scores by systematically combining buffer state, slice-specific QoS factors, spectral efficiency, and service priorities to adjust resource allocation priorities and minimize SLA violations objective in real time.
- 3) It proposes a practical three-phase resource allocation algorithm (i.e., minimum guarantee allocation (QoS-driven RBG estimation), weighted urgency-based distribution (computing a balanced share of slice-specific urgency and number of users), and priority-based RR allocation) (Section IV-C) that seamlessly balances immediate connectivity requirements with long-term SLA fulfillment objectives while maximizing spectrum utilization efficiency.
- 4) It demonstrates superior performance when compared to state-of-the-art alternatives using realistic channel conditions in three deployment scenarios via comprehensive trace-driven simulations (Section V-B). Specifically, AQPS achieves exceptional SLA assurance rates of 99.07% and 97.58% in Scenarios 1 and 2, respectively.

The rest of the article is structured as follows. Section II reviews recent optimization- and data-driven approaches to the network slicing-based RRA problem. Section III presents the AQPS system architecture and problem formulation. Section IV details the core algorithmic components and proposed RRA allocation mechanism. Section V describes the experimental setup and presents comprehensive performance evaluations for multiservice deployments. Finally, Section VI concludes the article and discusses future research directions.

II. RELATED WORKS

A. Optimization-Based RRA Approaches

Several RRA approaches have recently been proposed to ensure SLA fulfillment for different network slices. In this context, Campos et al. [23] proposed a stepwise optimal interslices RRA algorithm called SOA for SLA assurance in

5G networks. The authors formalized the stepwise interslices resource scheduling problem and developed a polynomial-time optimal algorithm by considering eMBB, URLLC, and BE slices. Chen et al. [32] introduced a channel-aware RAN slicing solution called RadioSaber that considers both interslice and enterprise scheduling through greedy algorithms to optimize spectrum efficiency. The solution resolves the cyclic dependency between schedulers and ensures fairness and slice isolation by utilizing channel quality information at both levels. However, a major limitation of this work is the fixed slice weights for resource allocation instead of dynamically adjusting them according to network conditions.

D'Oro et al. [33] developed a distributed RAN slicing framework with reduced complexity by decomposing the resource allocation problem into multiple subproblems that can be solved independently. While their approach achieves near-optimal performance with less computational overhead, it mainly emphasizes spectral efficiency and lacks full SLA compliance measures. Ksentini and Nikaein [27] proposed a framework for enforcing RAN network slicing by introducing a two-level scheduler with a radio slice allocator handling interslice scheduling and an intraslice scheduler managing resource distribution within each slice. The proposed approach works well for basic slice differentiation but does not adapt well to changing network conditions. Aijaz [28] proposed HAP-SliceR, a radio resource slicing framework designed specifically for haptic communications in 5G networks. This framework employs a hierarchical structure with auction mechanisms for resource allocation. Its narrow focus on haptic communications limits how widely it can be applied to general slicing scenarios.

B. Data-Driven Approaches

Lotfi et al. [29] proposed an attention-based DRL RB scheduling method in an O-RAN-compatible architecture. The authors implemented a value-attention network to optimize resource allocation for eMBB, URLLC, and mMTC slices. However, the proposed solution mainly focuses on individual RB allocation rather than RBG allocation. Nahum et al. [24] proposed an intent-aware (SLA-focused) RRA mechanism based on the SAC DRL model to systematically determine intent drifts and dynamic allocation of resources based on slice priorities and network conditions. However, the proposed DRL solution could severely violate the SLA requirements under fluctuating channel conditions. Similarly, Polese et al. [30] proposed an ORAN-compliant RRA solution using PPO for both interslice and intraslice resource allocation for eMBB, URLLC, and mMTC slice types. The proposed approach incorporates dual reward functions to maximize overall system performance metrics. However, the proposed implementation optimizes general system metrics rather than specific QoS requirements.

Mei et al. [31] combined DDPG and DQN RL methods to manage resource allocation and power control in a RAN slicing environment. The authors proposed a hierarchical two-level architecture with an upper controller to establish rate boundaries and a lower controller for detailed RB and power allocation among user groups. However, the

potential complexity in coordinating between the two control levels is a significant challenge while maintaining optimal performance across all service requirements. Antonius [35] proposed an innovative CNN-based game theory solution for network slice management in next-generation networks. The proposed framework contains an organizational architecture in which local RRM obtain subchannel allocations from slice-level RRM before assigning resources to UEs. Qiao et al. [15] proposed M3O-RANS, a multicell, multidimensional, and multitimescale O-RAN slicing framework to resolve resource allocation inefficiencies across heterogeneous 5G networks. Sciancalepore et al. [36] proposed RL-NSB, a 5G network slice broker that uses RL, which includes: 1) analysis of traffic and user mobility patterns; 2) a forecasting and learning framework tailored for each slice; 3) admission control optimization using spatial and traffic insights; and 4) a reinforcement mechanism to steer the system toward optimal configurations. The proposed approach achieves promising results in dynamic environments but may require long training periods before achieving optimal performance. Gharehgoli et al. [34] introduced maximizing infrastructure provider utility with DRL in an end-to-end network slicing setup when deployed under uncertain demands and channel states. The authors focused on eMBB, URLLC, and mMTC services and formulated a nonconvex mixed-integer problem. They employed an RDPG algorithm. However, the proposed memory-based DRL approach demonstrates good convergence properties and scalability but does not offer formal guarantees for meeting SLAs.

Table I shows a comparative analysis of AQPS and existing RRA approaches for network slicing. Although existing approaches have made significant contributions to RRA in network slicing, they exhibit two major limitations: 1) most solutions [27], [29], [30], [31], [32], [36] cannot work in an SLA-aware system as they do not explicitly consider specific QoS requirements for different slice types; and 2) many approaches [24], [29], [30], [31], [32], [34] typically allocate 100% of available RBGs regardless of the actual demand. In contrast, the proposed AQPS solution dynamically adapts to the unique requirements of each network slice and implements a flexible multiphase resource allocation strategy combining QoS-driven RBG estimation with dynamic urgency-based prioritization. Specifically, AQPS integrates buffer state monitoring, slice-specific QoS factors, spectral efficiency considerations, and service priority mechanisms to achieve precise resource allocation across heterogeneous services. The proposed AQPS solution achieves superior SLA assurance rates of 99.07% and 97.58% in IoT-dense urban and enhanced IoT coverage zone scenarios compared to DRL [24], RR [37], and SOA [23] approaches.

III. PROPOSED SYSTEM ARCHITECTURE AND PROBLEM FORMULATION

A. System Architecture

Fig. 1 presents the end-to-end hierarchical system architecture, which enables the distribution of radio resources to slice users with distinct priority classes. The proposed

TABLE I
COMPARATIVE ANALYSIS OF AQPS AND EXISTING RRA APPROACHES FOR NETWORK SLICING

Reference	Approach	Slices	SLA Aware	Dynamic Adapt.	Priority Based	Demand Aware
Ksentini & Nikaein [27]	Two-level scheduler	xMBB, URLLC, mMTC	✗	✓	✗	✗
Aijaz [28]	Hierarchical auction	Haptic	✗	✓	✗	✓
D'Oro et al. [33]	Distributed optimization	Generic	✗	✓	✗	✓
Campos et al. [23]	Stepwise optimal (SOA)	eMBB, URLLC, BE	✓	✓	✗	✓
Mei et al. [31]	DDPG+DQN hierarchical	eMBB, URLLC, mMTC	✗	✓	✓	✗
Polese et al. [30]	PPO-based O-RAN	eMBB, URLLC, mMTC	✗	✓	✗	✗
Lotfi et al. [29]	Attention-based DRL	eMBB, URLLC, mMTC	✗	✓	✗	✗
Gharehgoi et al. [34]	RDPG memory-based	eMBB, URLLC, mMTC	✗	✓	✗	✓
Nahum et al. [24]	SAC-based intent-aware	eMBB, URLLC, BE	✓	✓	✓	✗
Antonius [35]	CNN-Game theory	Generic	✗	✓	✗	✗
Qiao et al. [15]	Hierarchical learning	eMBB, URLLC, mMTC	✓	✓	✓	✗
AQPS (Proposed)	QoS-driven urgency-based prioritization approach	eMBB, URLLC, BE	✓	✓	✓	✓

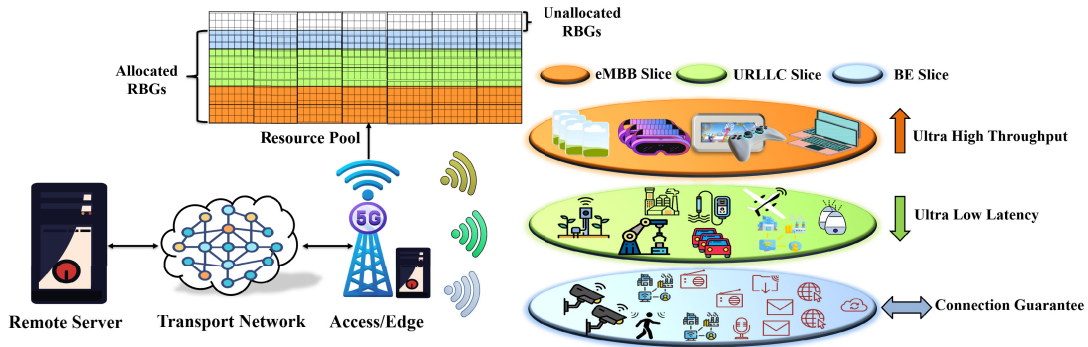


Fig. 1. AQPS system architecture to enable QoS-aware resource allocation within a 5G base station.

AQPS architecture includes a 5G base station with advanced network slice capabilities to orchestrate radio resources $\mathcal{R}$ across heterogeneous service requirements through three distinct network slices $\mathcal{S} = \{s^{\text{embb}}, s^{\text{urllc}}, s^{\text{be}}\}$. Each slice $s \in \mathcal{S}$ serves a designated set of users $\mathcal{U}_s$, where user associations are characterized by slice-specific priority classes: $\mathcal{U}_{s^{\text{urllc}}}$ for URLLC (Priority Class 1), $\mathcal{U}_{s^{\text{embb}}}$ for eMBB (Priority Class 2), and $\mathcal{U}_{s^{\text{be}}}$ for BE (Priority Class 3) traffic. The AQPS framework operates under strictly defined SLA requirements, where $\mathcal{T}_r^s$, $\mathcal{L}_r^s$, and $\mathcal{P}_r^s$ denote the throughput, latency, and packet loss rate requirements, respectively, for $s \in \{s^{\text{urllc}}, s^{\text{embb}}\}$, while $\mathcal{G}_r^{\text{be}}$ and $\mathcal{F}_r^{\text{be}}$ represent the long-term throughput and fifth-percentile throughput requirements for the BE slice. The system implements a time-slotted resource allocation mechanism where user-specific spectral efficiency values characterize channel conditions. Let $\eta_u(t)$ represent the spectral efficiency of user $u \in \mathcal{U}_s$ at time t . These values are derived from high-fidelity channel measurements using the QuaDRiGa¹ [24]. During each TTI $t \in \{1, 2, \dots, T\}$, the AQPS framework calculates the optimal number of RBGs allocated $\mathcal{A}_s(t)$ for all slices $s \in \mathcal{S}$. These resources are then distributed among users using an RR intraslice scheduling mechanism to maintain SLA compliance while optimizing resource utilization efficiency [38].

B. Problem Formulation

RRA in a 5G network environment is a multifaceted optimization challenge, particularly when addressing the heterogeneous service requirements of various connected users [39], [40]. The proposed AQPS system intelligently distributes limited radio resources across diverse network slices while meeting specific QoS constraints to overcome this challenge. The primary focus of AQPS is to minimize total SLA violations for all network slices over time. SLA violations for slice s at time t are denoted as $\mathcal{V}_s(t)$, and the comprehensive objective function is formulated as

$$\min \sum_{t=1}^T \sum_{s \in \mathcal{S}} \mathcal{V}_s(t). \quad (1)$$

The violation metric $\mathcal{V}_s(t)$ represents the deviations from the respective QoS requirements for each slice type, and is defined as

$$\mathcal{V}_s(t) = \begin{cases} \mathcal{V}_{s^{\text{urllc}}}(t), & \text{if } s = s^{\text{urllc}} \\ \mathcal{V}_{s^{\text{embb}}}(t), & \text{if } s = s^{\text{embb}} \\ \mathcal{V}_{s^{\text{be}}}(t), & \text{if } s = s^{\text{be}}. \end{cases} \quad (2)$$

Binary indicator functions that track whether specific QoS metrics exceed or fall below their respective requirements are used for slice-specific SLA violation capture. For URLLC and eMBB slices ($s \in \{s^{\text{urllc}}, s^{\text{embb}}\}$), violations are recorded when throughput, latency, or packet loss requirements are

¹<https://quadriga-channel-model.de/>

not satisfied. The violation metrics $\mathcal{V}_{\text{urllc}}(t)$ and $\mathcal{V}_{\text{embb}}(t)$ are computed as follows:

$$\mathcal{V}_s(t) = \sum_{u \in \mathcal{U}_s} \left(\mathbb{I}_{\{\Theta_u(t) < \mathcal{T}_r^s\}} + \mathbb{I}_{\{L_u(t) > \mathcal{L}_r^s\}} + \mathbb{I}_{\{P_u(t) > \mathcal{P}_r^s\}} \right) \quad (3)$$

where $L_u(t)$, $P_u(t)$, and $\Theta_u(t)$ represent the experienced latency, packet loss rate, and throughput for user u at time t , respectively. The indicator function $\mathbb{I}_{\{\cdot\}}$ equals 1 when the condition is true and 0 otherwise. The BE slice violations metric mainly considers long-term guarantees and minimum performance levels and is represented by

$$\mathcal{V}_{s^{\text{be}}}(t) = \sum_{u \in \mathcal{U}_{s^{\text{be}}}} \left(\mathbb{I}_{\{\Theta_u^{\text{agg}}(t) < \mathcal{G}_{r^{\text{be}}}^{\text{be}} \cdot w(t)\}} + \mathbb{I}_{\{\Theta_u(t) < \mathcal{F}_{r^{\text{be}}}^{\text{be}}\}} \right) \quad (4)$$

where $w(t)$ is the observation window size at time t , and $\Theta_u^{\text{agg}}(t)$ represents the aggregated throughput achieved by the user u over the previous $w(t) - 1$ intervals.

Constraints:

$$\sum_{s \in \mathcal{S}} \mathcal{A}_s(t) \leq \mathcal{R} \quad \forall t \quad (5)$$

$$\mathcal{A}_s(t) \geq 0 \quad \forall s \in \mathcal{S} \quad \forall t \quad (6)$$

$$\sum_{u \in \mathcal{U}_s^{\text{active}}(t)} \mathcal{A}_u(t) \geq \mathcal{A}_s^{\text{mg}}(t) \quad \forall s \in \mathcal{S} \quad \forall t \quad (7)$$

$$\sum_{u \in \mathcal{U}_s} \mathcal{A}_u(t) = \mathcal{A}_s(t) \quad \forall s \in \mathcal{S} \quad \forall t. \quad (8)$$

These constraints ensure the physical reliability and fairness of the resource allocation, where the constraint defined in (5) ensures that the sum of resources allocated to all slices cannot exceed the total available resources $\mathcal{R}$ in the system. The constraint in (6) is the non-negative resource allocation constraint that ensures the resource allocations to each slice must be positive. Constraint (7) ensures that the total RBGs allocated to users in each slice must meet or exceed the minimum guarantee allocation ($\mathcal{A}_s^{\text{mg}}(t)$), which is computed according to the hierarchical priority-based allocation (19)–(21) using QoS-driven resource requirements. This enables resource allocation to URLLC, eMBB, and BE users, in this order. The constraint in (8) ensures that the sum of the resources allocated to individual users ($\mathcal{A}_u(t)$) within a slice equals the total resources allocated to that slice.

This resource allocation optimization problem is NP-hard due to the complex inter-relationships between QoS metrics across different service types. To address this challenge, the proposed AQPS solution employs a priority-aware, urgency-driven heuristic approach to minimize the risk of potential SLA violations and hence enables informed and dynamic resource allocation decisions without the need for an exhaustive solution. The multiphase allocation approach described in Section IV-C first guarantees minimum connectivity for eMBB, URLLC, and BE users. It then distributes additional resources based on a weighted combination of slice urgency and active user proportion. Finally, it implements a protection mechanism for prioritized services. The proposed solution-oriented design balances computational efficiency with near-optimal performance while minimizing SLA violations across heterogeneous network slices. Next, we discuss the proposed framework

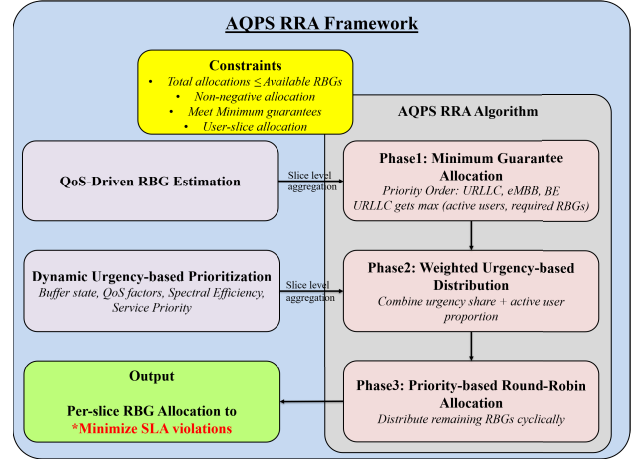


Fig. 2. Overall workflow of proposed AQPS RRA solution.

by highlighting how the QoS monitoring components interact with priority-based scheduling mechanisms to achieve real-time resource adaptation across the multislice architecture.

IV. PROPOSED AQPS RRA MODELING AND ALGORITHM

This section presents the system modeling and the proposed AQPS RRA algorithm. Fig. 2 illustrates the overall workflow of the AQPS framework, which consists of two key preprocessing stages and a three-phase resource allocation algorithm. Sections IV-A–IV-C detail each component of this framework.

A. QoS-Driven RBG Estimation

Effective radio resource distribution in IoT-enabled 5G network slicing environments demands an accurate assessment of resource needs based on varied QoS specifications. This is particularly important for IoT deployments where device density and traffic patterns vary significantly [41]. Here, we present a comprehensive framework for determining resource requirements across diverse slice categories and converting heterogeneous QoS metrics into standardized resource demand representations.

For URLLC users $u \in \mathcal{U}_{\text{urllc}}$, the required QoS, $\mathcal{Q}_u^{\text{req}}(t)$, is determined through

$$\mathcal{Q}_u^{\text{req}}(t) = \begin{cases} \max(\mathcal{Q}_u^{\text{min}}, \mathcal{Q}_u^{\text{buf}}(t)), & \text{if } \mathcal{B}_u^{\text{risk}}(t) > 0 \\ \mathcal{Q}_u^{\text{min}}, & \text{if } \mathcal{B}_u^{\text{risk}}(t) = 0 \end{cases} \quad (9)$$

where $\mathcal{Q}_u^{\text{min}}$ is the minimum guaranteed throughput from SLA requirements for URLLC users, i.e., $\mathcal{T}_r^s$. $\mathcal{Q}_u^{\text{buf}}(t)$ is the buffer-derived QoS needed to meet latency constraints.

AQPS employs a proactive latency management approach by identifying risky packets close to violating their delay constraints. The buffer-based throughput $\mathcal{Q}_u^{\text{buf}}(t)$ required to satisfy these latency constraints is computed as follows:

$$\mathcal{Q}_u^{\text{buf}}(t) = \frac{\mathcal{W}_u^{\text{pkt}} \cdot \mathcal{B}_u^{\text{risk}}(t)}{\tau} \quad (10)$$

where $\mathcal{B}_u^{\text{risk}}(t)$ represents the number of packets at risk of violating latency requirements at time t and is given as $(\sum_{i=\mathcal{L}_r^s}^{\mathcal{L}_u^{\text{max}}} \mathcal{B}_u^i(t))$, where $\mathcal{B}_u^i(t)$ represents the number of packets

that have waited for i time intervals at time t and $\mathcal{L}_u^{\max}$ is the maximum latency for user u . In (10), $\mathcal{W}_u^{\text{pkt}}$ is the packet size and τ is the TTI duration.

The AQPS framework converts these QoS requirements $\mathcal{Q}_u^{\text{req}}(t)$ into resource demands, i.e., RBGs. The required number of RBGs is calculated by considering the spectral efficiency of the user and the capacity of each RBG and is represented as follows:

$$\mathcal{R}_u^{\text{req}}(t) = \left\lceil \frac{\mathcal{Q}_u^{\text{req}}(t)}{\eta_u(t) \cdot \beta \cdot \nu} \right\rceil \quad (11)$$

where β is the bandwidth per RB, and ν is the number of RBs per RBG. This QoS-driven estimation approach provides a unified framework for converting diverse requirements from heterogeneous services into standardized resource demands. By considering explicit throughput requirements and latency constraints, the AQPS approach can accurately determine the resource needs.

B. Dynamic Urgency-Based Prioritization

AQPS considers a sophisticated urgency calculation mechanism to prioritize resource allocation based on each user's dynamic state and specific QoS constraints. This urgency-based approach enables the scheduler (discussed in Section IV-C) to adapt to changing system states while addressing diverse service requirements. The urgency score $\mathcal{X}_u(t)$ for a user $u \in \mathcal{U}_s$ at time t is defined as follows:

$$\mathcal{X}_u(t) = \max(0, (\mathcal{X}_u^b(t) + \mathcal{X}_u^s(t)) \cdot \mathcal{X}_u^\eta(t) \cdot p_u). \quad (12)$$

This comprehensive urgency model includes four key components to systematically decide how resources will be allocated among slices: 1) buffer state contribution ($\mathcal{X}_u^b(t)$); 2) slice-specific QoS factors ($\mathcal{X}_u^s(t)$); 3) spectral efficiency ($\mathcal{X}_u^\eta(t)$); and 4) service priority factor (p_u). These components provide a balanced approach to prioritizing resource distribution in dynamic network environments.

1) *Buffer State Contribution*: The buffer state component reflects the immediate transmission needs of a user based on current buffer occupancy, and is computed as

$$\mathcal{X}_u^b(t) = \begin{cases} B_u(t), & \text{if } B_u(t) > 0 \\ 0, & \text{otherwise} \end{cases} \quad (13)$$

where $B_u(t)$ represents the current buffer occupancy at time t . The buffer state factor prioritizes users with larger data queues and ensures they receive resources sooner to prevent delays and buffer overflow. In this way, the buffer occupancy better reflects the users' resource needs for maintaining smooth traffic flow and long-term connectivity.

2) *Slice-Specific Urgency Factors*: The urgency factor for URLLC users denoted by $\mathcal{X}_u^{\text{urllc}}(t)$ primarily depends on latency satisfaction, and is provided as

$$\mathcal{X}_u^{\text{urllc}}(t) = \frac{L_u(t)}{\mathcal{L}_r^{\text{urllc}}} \quad (14)$$

where $L_u(t)$ is the current average buffer latency. The URLLC-specific urgency factor ensures that urgency increases as latency approaches the strict URLLC requirement. This

formulation allows critical URLLC transmissions to be prioritized. For eMBB users, the urgency factor $\mathcal{X}_u^{\text{embb}}(t)$ balances maintaining high throughput and meeting latency satisfaction, and is represented as

$$\mathcal{X}_u^{\text{embb}}(t) = \frac{L_u(t)}{\mathcal{L}_r^{\text{embb}}} + \frac{\mathcal{T}_r^{\text{embb}}}{T_u(t)} \quad (15)$$

where $T_u(t)$ is the current throughput of the eMBB user. The dual-factor approach ensures that eMBB users receive increased bandwidth and adequate latency transmission. Next, the BE-specific urgency function focusing on long-term throughput is provided in equation (16):

$$\mathcal{X}_u^{\text{be}}(t) = \begin{cases} \frac{\mathcal{G}_r^{\text{be}} \cdot w(t) - \Theta_u^{\text{agg}}(t)}{\mathcal{G}_r^{\text{be}} \cdot w(t)}, & \text{if } \Theta_u^{\text{agg}}(t) < \mathcal{G}_r^{\text{be}} \cdot w(t) \\ 0, & \text{otherwise.} \end{cases} \quad (16)$$

Equation (16) ensures that the BE users who are at risk of not meeting their targeted long-term throughput get higher priority to maintain fairness while still respecting the lower priority of BE traffic compared to URLLC and eMBB services.

3) *Spectral Efficiency Factor*: The urgency model defined in (12) includes a scaling factor based on spectral efficiency to promote efficient resource utilization. $\mathcal{X}_u^\eta(t)$ function improves the overall system throughput by enabling the allocation of resources to users (with better channel conditions) who can use them more efficiently

$$\mathcal{X}_u^\eta(t) = 1.0 + \min(\eta_u(t), 1.0). \quad (17)$$

The formulation in (17) enables users with poor channel conditions to maintain a reasonable share of network resources while still encouraging efficient spectrum utilization.

4) *Service-Specific Priority Coefficient*: The urgency factor $\mathcal{X}_u(t)$ is scaled by a service-specific priority multiplier that reflects the distinct requirements of different service types. The formulation is represented as

$$p_u = \begin{cases} \psi_{s^{\text{urllc}}}, & \text{if } u \in \mathcal{U}_{s^{\text{urllc}}} \\ \psi_{s^{\text{embb}}}, & \text{if } u \in \mathcal{U}_{s^{\text{embb}}} \\ \psi_{s^{\text{be}}}, & \text{if } u \in \mathcal{U}_{s^{\text{be}}}. \end{cases} \quad (18)$$

These service-specific priority coefficients ($\psi_{s^{\text{urllc}}}$, $\psi_{s^{\text{embb}}}$, and $\psi_{s^{\text{be}}}$) enable the AQPS framework to be configurable according to specific deployment scenarios and operational requirements. In addition, this prioritization approach allows the implementation of critical service levels for high-priority traffic while still providing acceptable performance for lower priority services based on network conditions.

The proposed urgency model is employed in the AQPS resource allocation algorithm (Section IV-C) to comprehensively consider the resource allocation priorities of each user based on their current state, QoS requirements, and network conditions. These urgency scores ensure that resources are distributed in a manner that optimizes QoS satisfaction (i.e., SLA assurance) across all services while maintaining efficient resource utilization. The dynamic nature of the urgency formulation allows AQPS to adaptively respond to changing traffic patterns and network conditions, making it particularly suitable for heterogeneous and dynamic 5G IoT networks.

C. AQPS Resource Allocation Algorithm

Algorithm 1 AQPS: QoS- and Priority-Aware RRA Algorithm

Input: Set of slices $\mathcal{S} = \{s^{urllc}, s^{embb}, s^{be}\}$
 Set of users $\mathcal{U}_s$ for each slice s
 Number of limited RBGs $\mathcal{R}$
 Parameters: $w_{max}, \eta, \mathcal{B}, \nu, \tau, TTIs$
 Requirements: $\mathcal{T}_r^{urllc}, \mathcal{T}_r^{embb}, \mathcal{G}_r^{be}, \mathcal{F}_r^{be}$
Output: Resource allocation $\mathcal{A}_s(t)$ for all $s \in \mathcal{S}$

```

1 for  $t \leftarrow 1$  to  $TTIs$  do
2    $w(t) \leftarrow \min(t, w_{max})$ 
3   foreach  $s \in \mathcal{S}$  do
4      $\mathcal{U}_s^{active}(t) \leftarrow \{u \in \mathcal{U}_s : B_u(t) > 0 \text{ or } Q_u^{min} > 0\}$ 
5     foreach  $u \in \mathcal{U}_s^{active}(t)$  do
6       Calculate  $Q_u^{req}(t)$  using (9-10)
7       Calculate  $\mathcal{R}_u^{req}(t)$  using (11)
8       Calculate  $\mathcal{X}_u(t)$  with parameter  $w(t)$  using (12-18)
9     end
10    Calculate  $\mathcal{R}_s^{req}(t) = \sum_{u \in \mathcal{U}_s^{active}(t)} \mathcal{R}_u^{req}(t)$ 
11    Calculate  $\mathcal{X}_s(t) = \sum_{u \in \mathcal{U}_s^{active}(t)} \mathcal{X}_u(t)$ 
12    Compute minimum allocation  $\mathcal{A}_s^{mg}(t)$  using (19-21)
13     $\mathcal{R}^{mg}(t) = \mathcal{R} - \sum_{s \in \mathcal{S}} \mathcal{A}_s^{mg}(t)$ 
14    Calculate weighted urgency factor  $\mathcal{X}_s^w(t)$  using (22)
15    Calculate resource share  $\mathcal{C}_s(t)$  using (23)
16    Compute weighted urgency-based allocation  $\mathcal{A}_s^{wu}(t)$  using (24)
17     $\mathcal{R}^{wu}(t) = \mathcal{R}^{mg}(t) - \sum_{s \in \mathcal{S}} \mathcal{A}_s^{wu}(t)$ 
18    Compute priority-based allocation  $\mathcal{A}_s^{pb}(t)$  using Algorithm 2
19  end
20  return  $\mathcal{A}_s(t)$  for all  $s \in \mathcal{S}$ 
21 end

```

The AQPS framework employs a multiphase resource allocation approach to efficiently distribute RBGs among network slices while satisfying their QoS needs. Algorithm 1 presents the proposed AQPS resource allocation mechanism that operates on a per-TTI basis. The sliding window is set for each decision interval to the minimum of t and w_{max} (line 2). The algorithm then computes the active users $\mathcal{U}_s^{active}(t)$ for each slice (line 4). Users with no throughput requirement or buffer levels are considered inactive for different slices. Next, the algorithm computes the required QoS and RBGs described in Section IV-A (lines 6–7). In parallel, the algorithm calculates the user urgency requirements $\mathcal{X}_u(t)$, as defined in Section IV-B (line 8). The RBGs and urgency demands of users are then aggregated to reflect the RBGs and urgency requirements per slice and are denoted as $\mathcal{R}_s^{req}(t)$ and $\mathcal{X}_s(t)$, respectively (lines 10–11). The algorithm then proceeds with resource allocations to slices $\mathcal{A}_s(t)$ to ensure best-fit connectivity for all users. The allocation process consists of four sequential phases, which are described below.

1) *Minimum Guarantee Allocation:* The proposed solution first allocates the minimum necessary RBGs to the URLLC slice for minimum guarantee allocation due to its stringent latency requirements

$$\mathcal{A}_{s^{urllc}}^{mg}(t) = \min(\max(|\mathcal{U}_{s^{urllc}}^{active}(t)|, \mathcal{R}_{s^{urllc}}^{req}(t)), \mathcal{R}). \quad (19)$$

In the first allocation phase, the $\mathcal{A}_{s^{urllc}}^{mg}(t)$ guarantees that the URLLC slice receives either enough RBGs to satisfy all its users' requirements ($\mathcal{R}_{s^{urllc}}^{req}(t)$) or at least one RBG per active user ($|\mathcal{U}_{s^{urllc}}^{active}(t)|$), subject to the constraint of total available resources ($\mathcal{R}$). After allocating resources to the URLLC slice, the remaining resources $\mathcal{R} - \mathcal{A}_{s^{urllc}}^{mg}(t)$ are available for the eMBB slice. For the eMBB slice, the minimum guarantee allocation is computed as follows:

$$\mathcal{A}_{s^{embb}}^{mg}(t) = \min(|\mathcal{U}_{s^{embb}}^{active}(t)|, \mathcal{R} - \mathcal{A}_{s^{urllc}}^{mg}(t)). \quad (20)$$

The allocation $\mathcal{A}_{s^{embb}}^{mg}(t)$ ensures that each active eMBB user receives at least one RBG if sufficient resources remain after the URLLC allocation. Finally, similar to the eMBB allocations, the minimum remaining resources are allocated to the BE slice

$$\mathcal{A}_{s^{be}}^{mg}(t) = \min(|\mathcal{U}_{s^{be}}^{active}(t)|, \mathcal{R} - \mathcal{A}_{s^{urllc}}^{mg}(t) - \mathcal{A}_{s^{embb}}^{mg}(t)). \quad (21)$$

The first allocation phase maintains the priority hierarchy (i.e., URLLC, eMBB, and BE) while ensuring each slice receives at least its minimum required resources whenever possible, subject to the constraint of total available RBGs (line 12).

2) *Weighted Urgency-Based Distribution:* In the second allocation phase, the remaining RBGs $\mathcal{R}^{mg}(t)$ (computed on line 13 of Algorithm 1) after the minimum guarantee allocation are distributed based on a weighted combination of slice urgency and active users proportion. For each slice $s \in \mathcal{S}$ that has active users $\mathcal{U}_s^{active}(t)$, AQPS calculates weighted urgency (line 14) using the slice's total urgency metric $\mathcal{X}_s(t)$ and a priority factor ψ_s specific to that slice

$$\mathcal{X}_s^w(t) = \mathcal{X}_s(t) \cdot \psi_s \quad (22)$$

where $\mathcal{X}_s(t)$ captures the slice's QoS requirements, and ψ_s reflects the relative importance of different slice types in the network.

Next, we compute a combined resource share (line 15) for each slice that balances urgency and active user proportions in the equation (23):

$$\mathcal{C}_s(t) = \alpha_s \cdot \frac{\mathcal{X}_s^w(t)}{\sum_{s' \in \mathcal{S}} \mathcal{X}_{s'}^w(t)} + (1 - \alpha_s) \cdot \frac{|\mathcal{U}_s^{active}(t)|}{\sum_{s' \in \mathcal{S}} |\mathcal{U}_{s'}^{active}(t)|} \quad (23)$$

where the first term represents the slice's proportion of the total weighted urgency across all slices, and the second represents the slice's proportion of total active users. The slice-specific balance parameters ($\alpha_{s^{urllc}}$, $\alpha_{s^{embb}}$, and $\alpha_{s^{be}}$) determine the relative influence of urgency metrics versus active user proportions for each slice type. Higher values of α_s increase the impact of urgency metrics, while lower values give more weight to the proportion of active users within the slice. Finally, the weighted urgency-based RBG allocation (line 16) for each slice is defined as follows:

$$\mathcal{A}_s^{wu}(t) = \lfloor \mathcal{R}^{mg}(t) \cdot \mathcal{C}_s(t) \rfloor. \quad (24)$$

The floor function in (24) ensures integer RBG allocation during the weighted urgency phase. The remaining RBGs are systematically distributed through the priority-based RR mechanism.

Algorithm 2 Priority-Based RR Allocation

Input: $\mathcal{R}^{wu}(t)$: Remaining RBGs after weighted urgency allocation

Output: $\mathcal{A}_s^{pb}(t)$ for each slice s

```

1  $\mathcal{O}(t) \leftarrow \emptyset$ 
2 for  $s \in \{s^{urllc}, s^{embb}, s^{sbe}\}$  do
3   if  $|\mathcal{U}_s^{active}(t)| > 0$  then
4      $\mathcal{O}(t) \leftarrow \mathcal{O}(t) \cup \{s\}$ 
5   end
6 end
7 if  $|\mathcal{O}(t)| > 0$  then
8    $i \leftarrow 0$ 
9   for  $s \in \mathcal{S}$  do
10     $\mathcal{A}_s^{pb}(t) \leftarrow 0$ 
11  end
12  while  $i < \mathcal{R}^{wu}(t)$  do
13     $s \leftarrow \mathcal{O}(t)[i \bmod |\mathcal{O}(t)|]$ 
14     $\mathcal{A}_s^{pb}(t) \leftarrow \mathcal{A}_s^{pb}(t) + 1$ 
15     $i \leftarrow i + 1$ 
16  end
17 end
18 return  $\mathcal{A}_s^{pb}(t)$  for each slice  $s$ 
```

3) *Priority-Based RR Distribution*: After the weighted urgency-based allocation, we implement a priority-based RR distribution mechanism for any remaining RBGs. Let $\mathcal{R}^{wu}(t)$ represent the unallocated RBGs after the weighted urgency phase (Algorithm 1, line 17). Algorithm 1 then computes the priority-based allocations $\mathcal{A}_s^{pb}(t)$ for each slice type. The priority-based RR distribution mechanism is presented in Algorithm 2. This allocation ensures complete resource utilization while maintaining the QoS proportions among slices. The algorithm first creates a sequence $\mathcal{O}(t)$ of slices with active users in the order of inherent priority levels, i.e., s^{urllc} , s^{embb} , and s^{sbe} (Algorithm 2, lines 2–6). Only slices with at least one active user at time t are included in this ordered set. The algorithm then systematically distributes the remaining RBGs by iterating through $\mathcal{O}(t)$ in a cyclic fashion (Algorithm 2, lines 7–17), allocating exactly one RBG to each slice per iteration until all remaining resources are exhausted. The low-complex deterministic RR approach ensures that when resources cannot be divided equally among slices, the higher priority slices receive preferential treatment and are allocated additional resources. For each slice, the final priority-based allocation $\mathcal{A}_s^{pb}(t)$ returns the total number of RBGs distributed to the slice s during this RR allocation phase (Algorithm 2, line 13). The final allocation (line 20) for each slice at time t is computed as the sum of allocations from all three phases in equation (25):

$$\mathcal{A}_s(t) = \mathcal{A}_s^{mg}(t) + \mathcal{A}_s^{wu}(t) + \mathcal{A}_s^{pb}(t). \quad (25)$$

The resource allocation algorithm ensures that the resource allocation never exceeds the available resources, i.e.,

$\sum_{s \in \mathcal{S}} \mathcal{A}_s(t) \leq \mathcal{R}$. The computational complexity of AQPS is $\mathcal{O}(|\mathcal{S}| \cdot |\mathcal{U}|)$, where $|\mathcal{S}|$ is the number of slices. This dynamic multiphase allocation in AQPS supports all slices with basic connectivity through minimum guarantee allocation while balancing QoS across different numbers of active users. Distributing any remaining resources based on slice preferences in IoT transmission environments enhances overall resource utilization [42].

V. EXPERIMENTAL SETTINGS AND EVALUATION

A. AQPS Resource Allocation Environment

The proposed AQPS solution is simulated using a Python-based network slicing simulator that implements two IoT environments: an IoT-dense urban scenario (Scenario 1) and an enhanced IoT coverage zone (Scenario 2). The system architecture comprises a base station with the support of eMBB, URLLC, and BE network slices, each configured with requirements for two scenarios. For the urban scenario (Scenario 1), we set the base station to 100 MHz bandwidth with $\mathcal{R} = 138$ RBGs, each containing $\nu = 4$ RBs. Each RB uses a 180-kHz bandwidth (β), in line with 5G NR specs [43]. The TTI duration τ is 1 ms, using a window size $w_{\max}$ of 10 timestamps. In the coverage zone (Scenario 2), we used a 150-MHz bandwidth with 104 total RBGs per allocation cycle. Scenario 2 RBs have 360-kHz bandwidth (β). We tested both scenarios with limited RBGs: $\mathcal{R} = 80$. Scenario 2 uses a shorter TTI duration τ of 0.5 ms while keeping the same window size $w_{\max}$ of 10 timestamps.

Channel conditions are modeled through user-specific spectral efficiency values derived from the QuaDRiGa channel simulator [44], which incorporates comprehensive path loss models, shadowing effects, and intercell interference for high-demand scenarios. The simulator implements realistic channel conditions by considering the dual-slope path loss statistical models of 3GPP 38.901 UMi and includes interference from six neighboring cells [45], [46].

The realistic experimental settings, including user distribution and system parameters, are similar to those found in Campos et al. [23] and Nahum et al. [24]. In both scenarios, we considered consistent user distribution across slices, with $|\mathcal{U}_{s^{embb}}| = 4$, $|\mathcal{U}_{s^{urllc}}| = 3$, and $|\mathcal{U}_{s^{sbe}}| = 3$ users as also used in [24]. The slice-specific priority weights for both scenarios are set at: $\alpha_{s^{urllc}} = 0.8$, $\alpha_{s^{embb}} = 0.5$, and $\alpha_{s^{sbe}} = 0.2$. Similarly, weighting factors γ_b , γ_l , γ_t , γ_g , and γ_η are set at 0.5 to provide balanced consideration of slice-specific urgency factors in dynamic resource allocation decisions. To maintain the relative importance of each slice, we configured the service-specific priority parameters for both IoT scenarios after careful testing with $\psi_{s^{urllc}} = 0.545$, $\psi_{s^{embb}} = 0.273$, and $\psi_{s^{sbe}} = 0.182$. In both testing scenarios, the simulation duration is 2000 TTIs.

For both scenarios, we considered distinct SLA requirements for each slice. For Scenario 1, the eMBB slice requirements include: $\mathcal{T}_r^{s^{embb}} = 15$ Mbps, $\mathcal{L}_r^{s^{embb}} = 15$ ms, and $\mathcal{P}_r^{s^{embb}} = 20\%$, with a traffic rate of 25 Mbps, user latency configuration of 100 ms, packet size of 1500 bytes, and buffer size of 64 kB. The URLLC slice has stringent requirements: $\mathcal{T}_r^{s^{urllc}} = 2$ Mbps, $\mathcal{L}_r^{s^{urllc}} = 1$ ms, and $\mathcal{P}_r^{s^{urllc}} = 0.0001\%$,

TABLE II
AQPS RRA SYSTEM PARAMETERS AND QoS CONFIGURATIONS FOR SCENARIOS 1 AND 2

Configuration	Scenario 1	Scenario 2
System Parameters		
Bandwidth	100 MHz	150 MHz
Sub carrier width	15 KHz	30 KHz
Resource Blocks	555	416
Total / Used Resource Block Groups ($\mathcal{R}$)	138/80	104/80
RBs per RBG (ν) / RB bandwidth (β)	4 / 180 kHz	4 / 360 kHz
TTI duration (τ) / Time window (w_{max})	1 ms / 10 timestamps	0.5 ms / 10 timestamps
Users (eMBB, URLLC, BE)	4, 3, 3	4, 3, 3
$\alpha_{s_{urllc}}, \alpha_{s_{embb}}, \alpha_{s_{be}}$	0.8, 0.5, 0.2	0.8, 0.5, 0.2
$\gamma_b, \gamma_l, \gamma_t, \gamma_g, \gamma_\eta$	0.5	0.5
$\psi_{s_{urllc}}, \psi_{s_{embb}}, \psi_{s_{be}}$	0.545, 0.273, 0.182	0.545, 0.273, 0.182
Simulation duration	2000 TTIs	2000 TTIs
eMBB Slice Requirements		
Min. throughput / Traffic rate	15 Mbps / 25 Mbps	30 Mbps / 30 Mbps
Max. latency (Req. / User config)	15 ms / 100 ms	10 ms / 80 ms
Max. packet loss / Packet size / Buffer size	20% / 1500 bytes / 64 KB	10% / 1500 bytes / 128 KB
URLLC Slice Requirements		
Min. throughput / Traffic rate	2 Mbps / 2.5 Mbps	5 Mbps / 6 Mbps
Max. latency (Req. / User config)	1 ms / 50 ms	1 ms / 10 ms
Max. packet loss / Packet size / Buffer size	0.0001% / 250 bytes / 48 KB	0.00001% / 256 bytes / 32 KB
BE Slice Requirements		
Long-term / 5th-percentile throughput	5 Mbps / 3 Mbps	10 Mbps / 5 Mbps
Traffic rate / Max. latency (User config)	20 Mbps / 150 ms	20 Mbps / 200 ms
Packet size / Buffer size	1500 bytes / 64 KB	1500 bytes / 128 KB

with traffic rate of 2.5 Mbps. For BE slice we set the requirements of $\mathcal{G}_r^{s_{be}} = 5$ Mbps long-term throughput and $\mathcal{F}_r^{s_{be}} = 3$ Mbps fifth-percentile throughput. For Scenario 2, the eMBB slice requirements are enhanced to: $\mathcal{T}_r^{s_{embb}} = 30$ Mbps, $\mathcal{L}_r^{s_{embb}} = 10$ ms, and $\mathcal{P}_r^{s_{embb}} = 10\%$. The URLLC slice requirements are set to: $\mathcal{T}_r^{s_{urllc}} = 5$ Mbps, with unchanged $\mathcal{L}_r^{s_{urllc}} = 1$ ms, but more stringent $\mathcal{P}_r^{s_{urllc}} = 0.00001\%$. The BE slice requirements are increased to $\mathcal{G}_r^{s_{be}} = 10$ Mbps long-term throughput and $\mathcal{F}_r^{s_{be}} = 5$ Mbps fifth-percentile throughput. Table II presents the parameters and values for our three-slice network architecture with system-level radio resource settings and slice-specific parameters for eMBB, URLLC, and BE services.

The proposed AQPS solution dynamically adapts resource allocation based on instantaneous network conditions, slice requirements, and urgency factors while ensuring efficient radio resource utilization and SLA compliance. AQPS is compared with an off-policy SAC DRL [24], RR [37], and SOA [23] solutions using default settings.

B. Experimental Results

This subsection describes the resource allocation performance in resource-constrained scenarios for the proposed AQPS, DRL, RR, and SOA solutions across three network slices.

1) *EMBB, URLLC, and BE Slices Performance for Scenario 1:* Fig. 3 shows the QoS performance achieved by the WPU of the eMBB slice when deploying AQPS, DRL, RR, and SOA solutions. Fig. 3(a) demonstrates the robust performance of the AQPS solution, where almost 60% of throughput distribution exceeds 25 Mbps for WPU, and only about 18% falls below the critical 15 Mbps $\mathcal{T}_r^{s_{embb}}$ requirement. In contrast, DRL and RR solutions exhibit the poorest throughput

performance, with approximately 40% of measurements below the $\mathcal{T}_r^{s_{embb}}$ threshold. For latency $\mathcal{L}_r^{s_{embb}}$ fulfillment [Fig. 3(b)], AQPS significantly outperforms all comparative solutions and maintains 100% of latency measurements below the 15-ms threshold. SOA shows comparable latency performance to AQPS; however, AQPS maintains 80% of measurements below 10 ms, while SOA keeps almost 40% of the latency below 10 ms. Both DRL and RR display substantially higher latency values, i.e., 80% and 60% above 15 ms. The packet loss rate results in Fig. 3(c) illustrate the superior performance of AQPS and SOA solutions, with approximately 78% of measurements below 5% packet loss. Unlike AQPS, DRL and RR methods cannot keep acceptable PLR values, exceeding $\mathcal{P}_r^{s_{embb}}$ limits for over 15% of the test period. AQPS never goes above the 20% $\mathcal{P}_r^{s_{embb}}$ threshold, showing how well it maintains essential QoS metrics even for worst-performing eMBB users under resource-constrained network conditions.

Next, the throughput and latency values for URLLC users are shown in Fig. 3(d) and (e). It is interesting to note that all solutions have a zero packet loss rate for URLLC services. At the same time, AQPS delivers outstanding throughput performance for URLLC users with almost 100% of measurements exceeding the $\mathcal{T}_r^{s_{urllc}} = 2$ Mbps requirement, while outperforming SOA and DRL solutions, which fail to achieve SLA compliance. However, these methods show satisfactory performance, with 80% of measurements beating the requirement [Fig. 3(d)]. Fig. 3(e) illustrates the latency distribution for the worst-performing URLLC user. Here, AQPS demonstrates superior performance with nearly 90% of measurements below 0.5 ms and significantly outperforms SOA, which only has about 0% below this mark. DRL and RR solutions show good latency performance; however, the average latency values for all users with the DRL method exceed the AQPS solutions, as illustrated in Fig. 3(f). For BE traffic, AQPS

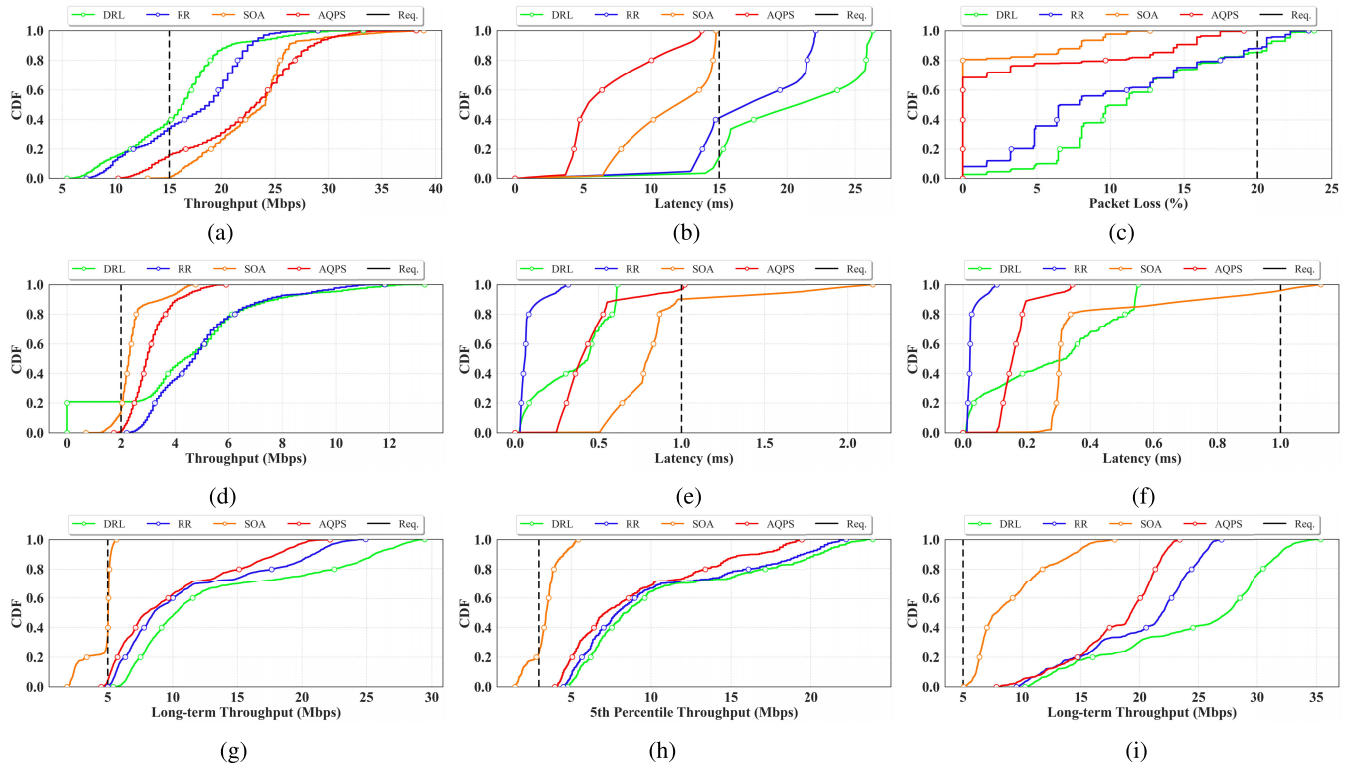


Fig. 3. Scenario 1: QoS performance values achieved by AQPS, DRL, RR, and SOA resource allocation solutions for WPU of eMBB, URLLC, and BE slices. (a) eMBB throughput (WPU). (b) eMBB latency (WPU). (c) eMBB packet loss rate (WPU). (d) URLLC throughput (WPU). (e) URLLC latency (WPU). (f) URLLC latency (Avg). (g) BE long-term throughput (WPU). (h) BE fifth-percentile throughput (WPU). (i) BE long-term throughput (Avg).

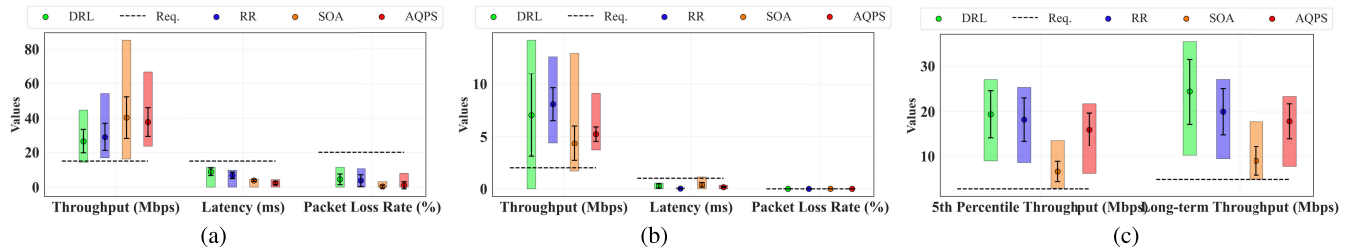


Fig. 4. Scenario 1: Average achieved QoS values for eMBB, URLLC, and BE slice users across AQPS, DRL, RR, and SOA resource allocation solutions. (a) eMBB-average achieved values. (b) URLLC-average achieved values. (c) BE-average achieved values.

finds a good balance between guaranteed minimums and making the most available resources when possible. The $\mathcal{G}_r^{\text{be}}$ performance of AQPS for BE slices consistently exceeds the 5-Mbps requirement throughout the simulation period [Fig. 3(g)], providing more than 16 Mbps on average, superior to SOA, which struggles to meet the requirement throughout the entire period. Similarly, $\mathcal{F}_r^{\text{be}}$ throughput measurements in Fig. 3(h) show AQPS maintains values well above the 3-Mbps threshold with stabilization around 20 Mbps. SOA is the only solution to experience substantial fluctuations, dropping as low as 3.0 Mbps during several intervals and consistently underperforming relative to the DRL and RR solutions. In contrast to the WPU, the average long-term throughput values in Fig. 3(i) for all solutions exceed the $\mathcal{G}_r^{\text{be}}$ requirement, with SOA being the lowest average solution.

The average achieved performance values across all slices for Scenario 1 in Fig. 4 illustrate that AQPS is a balanced approach to fulfill the heterogeneous QoS requirements in

multislice environments. For eMBB services, AQPS achieves an average throughput of 37.80 Mbps, compared to the requirement of 15 Mbps, with an average latency of 2.35 ms and minimal packet loss of 1.11% [Fig. 4(a)]. In URLLC environments, AQPS maintains exceptional performance with 0.17 ms average latency against the strict 1-ms requirement, compared to SOA's concerning 0.39 ms average latency that occasionally exceeds requirements [Fig. 4(b)]. Similarly, for BE traffic the fifth-percentile throughput value of 16.09 Mbps and the long-term throughput value of 18 Mbps for AQPS in Scenario 1 (Fig. 4(c)) are significant as they exceed the 3 Mbps ($\mathcal{F}_r^{\text{be}}$) and 5 Mbps ($\mathcal{G}_r^{\text{be}}$) requirements, respectively. This demonstrates AQPS's ability to maintain stable performance across different slices by effectively meeting the diverse and competing SLA requirements in 5G environments. It is important to note that the observed margins between SLA requirements and achieved average performance metrics reflect the effectiveness of our proposed AQPS solution. After

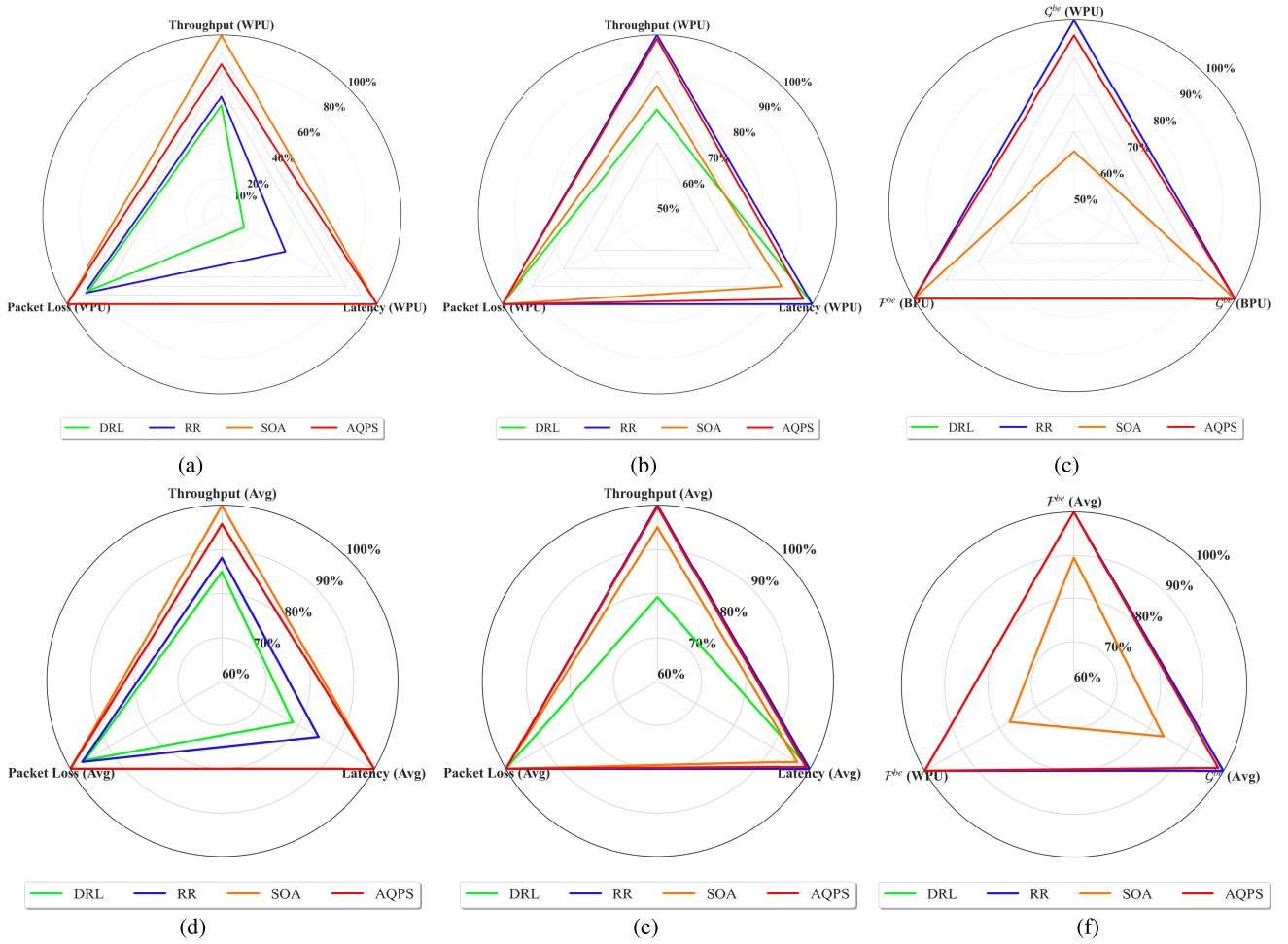


Fig. 5. Scenario 1: SLA assurance rates achieved by AQPS, DRL, RR, and SOA resource allocation solutions for different performing users of eMBB, URLLC, and BE slices. (a) eMBB SLA assurance (WPU). (b) URLLC SLA assurance (WPU). (c) BE SLA assurance (WPU & BPU). (d) eMBB SLA assurance (AAU). (e) URLLC SLA assurance (AAU). (f) BE SLA assurance (Avg & WPU).

TABLE III
COMPARATIVE SLA ASSURANCE PERFORMANCE OF AQPS, DRL, RR, AND SOA SOLUTIONS ACROSS TWO OPERATIONAL SCENARIOS

Method	Scenario 1			Scenario 2		
	Total Timestamps	Total Violations	SLA Assurance (%)	Total Timestamps	Total Violations	SLA Assurance (%)
DRL	3208	4472	91.72%	5371	7359	86.37%
RR	2102	2370	95.61%	2608	3590	93.35%
SOA	1644	2096	96.12%	1496	1847	96.58%
AQPS	485	500	99.07%	1141	1306	97.58%

minimum guarantee allocations, the weighted urgency-based and priority-based RR phases distribute remaining resources to improve average performance beyond minimums. This behavior is consistent with other SLA-aware resource allocation approaches, such as the one proposed by [24].

a) SLA Assurance in Scenario 1: The SLA assurance performance comparison of four resource allocation solutions for the eMBB, URLLC, and BE users is shown in Fig. 5. Fig. 5(a) and (b) provides SLA assurance results for the WPU for each method, while Fig. 5(d) and (e) provides SLA assurance values for the average of all users. Fig. 5(a) reveals that AQPS and SOA solutions achieve comparable maximum throughput performance for the WPU of eMBB services. At the same time, both solutions successfully fulfill the $\mathcal{L}_r^{\text{embb}}$ and

$\mathcal{P}_r^{\text{embb}}$ requirements for the eMBB slice. In contrast, DRL and RR solutions are the worst performing solutions and ensure almost 60% SLA compliance for $\mathcal{T}_r^{\text{embb}}$ with up to 20% $\mathcal{L}_r^{\text{embb}}$ assurance. Similar behavior can be observed in Fig. 5(d) where RR and DRL solutions exhibit notably reduced throughput and latency assurance. For packet loss metrics, AQPS and SOA deliver 100% requirement assurance, while RR shows acceptable performance of around 95%. In contrast to the WPU use-case in the eMBB slice, SOA struggles to attain consistent and higher throughput assurance for the URLLC slice [Fig. 5(b)]. RR is the best solution in terms of achieving maximum $\mathcal{T}_r^{\text{urllc}}$ and $\mathcal{L}_r^{\text{urllc}}$ satisfactions, closely followed by the proposed AQPS solution, which assures 100% $\mathcal{P}_r^{\text{urllc}}$ guarantees. A similar trend can be seen in Fig. 5(e), where

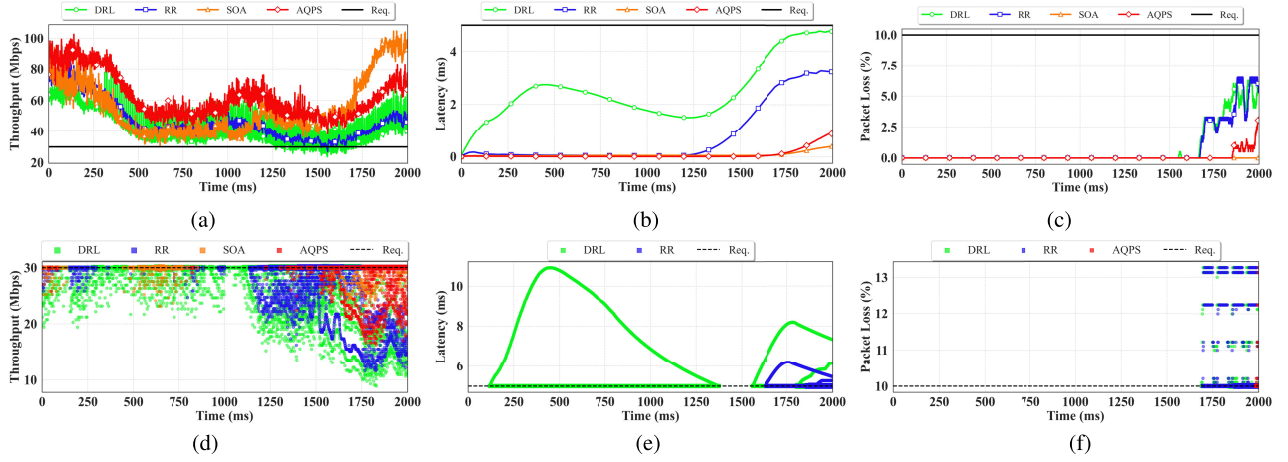


Fig. 6. Scenario 2: QoS performance metrics and SLA violation rates for eMBB slice users across AQPS, DRL, RR, and SOA resource allocation solutions. (a) eMBB throughput. (b) eMBB latency. (c) eMBB packet loss rate. (d) S2: T_r^{emb} violations. (e) S2: L_r^{emb} violations. (f) S2: P_r^{emb} violations.

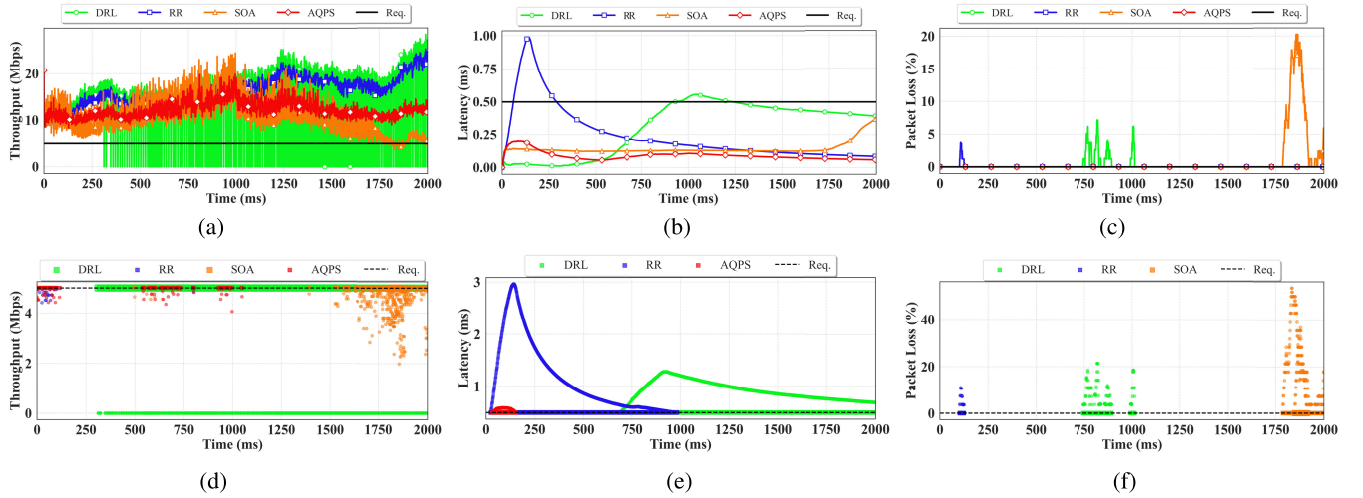


Fig. 7. Scenario 2: QoS performance metrics and SLA violation rates for URLLC slice users across AQPS, DRL, RR, and SOA resource allocation solutions. (a) URLLC throughput. (b) URLLC latency. (c) URLLC packet loss rate. (d) S2: T_r^{urllc} violations. (e) S2: L_r^{urllc} violations. (f) S2: P_r^{urllc} violations.

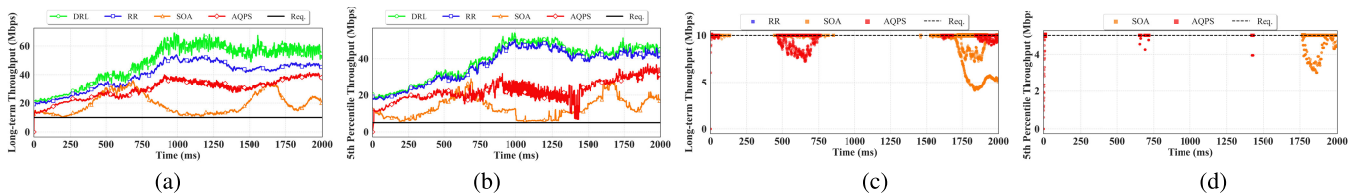


Fig. 8. Scenario 2: QoS performance metrics and SLA violation rates for BE slice users across AQPS, DRL, RR, and SOA resource allocation solutions. (a) BE long-term throughput. (b) BE fifth-percentile throughput. (c) S2: G_r^{be} violations. (d) S2: X_r^{be} violations.

the throughput and latency performance of the AQPS solution more closely match the RR solution. This is because the priority and urgency-based resource distribution in the AQPS solution ensures a useful tradeoff among different services. The DRL and RR solutions allocate most of the resources for BE services and achieve unnecessary top performance for nonurgent traffic applications, as seen in Fig. 5(c). However, AQPS maintains consistent flow to maintain G_r^{be} requirements for both WPU and BPU. SOA struggles to meet the G_r^{be} and X_r^{be} assurance for almost all users [Fig. 5(f)].

In summary, the AQPS solution demonstrates the most balanced SLA assurance performance across all three network slices in Scenario 1 and effectively manages the competing requirements of different service types compared to baseline solutions. This balanced resource allocation strategy enables AQPS to achieve an exceptional 99.07% SLA assurance rate compared to other solutions (Table III).

2) *EMBB, URLLC, and BE Slices Performance for Scenario 2:* Performance comparisons of four solutions for eMBB, URLLC, and BE slices for resource-constrained Scenario

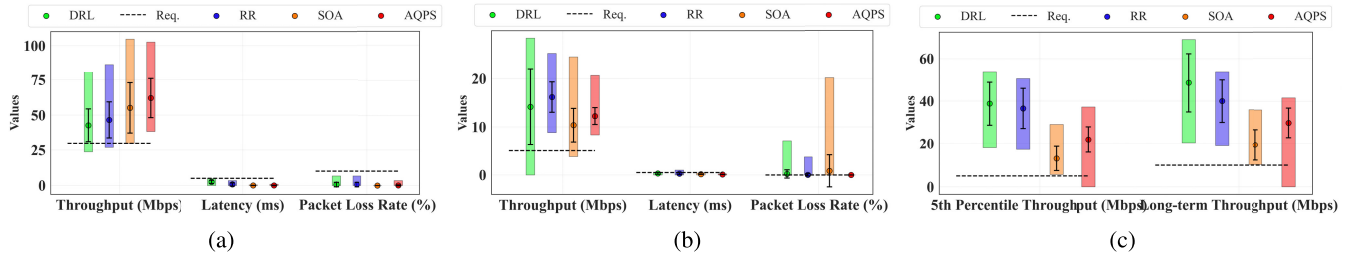


Fig. 9. Scenario 2: Average achieved QoS values for eMBB, URLLC, and BE slice users across AQPS, DRL, RR, and SOA resource allocation solutions. (a) eMBB-average achieved values. (b) URLLC-average achieved values. (c) BE-average achieved values.

2 are shown in Figs. 6–8. Compared to Scenario 1, we have almost doubled the SLA requirements in Scenario 2. The per-TTI throughput distribution in Fig. 6(a) shows that AQPS (with an average of 62.06 Mbps) and SOA (with an average of 55.16 Mbps) solutions maintain all users above the 30-Mbps requirement threshold. Both SOA and AQPS solutions exhibit the lowest $\mathcal{T}_r^{\text{embb}}$ violations with a percentage of 1.7% and 2.52%. Similarly, both AQPS and SOA solutions have the lowest latency values [Fig. 6(b)] for the eMBB slice and exceptionally fulfill the SLA requirements $\mathcal{L}_r^{\text{embb}}$ with no violations [Fig. 6(e)]. In contrast, the DRL solutions experience considerable latency fluctuations, peaking at approximately 4.5 ms after timestamp 1750, and RR, which shows gradually increasing latency, reaching 3.5 ms by simulation end.

In terms of latency, SOA initially demonstrates comparable performance to AQPS until around 1500 ms, but then quickly worsens, going beyond the critical 1-ms URLLC requirement. The packet loss performance in Fig. 6(c) shows how much better AQPS maintains broadband services with almost no packet loss for most of the simulation period and only minimal losses (about 2.5%) near the end. In comparison, DRL and RR exhibit steadily increasing packet loss rates beginning around 1700 ms, reaching about 7.0% at the end of the simulation. For the URLLC slice, AQPS consistently delivers $\mathcal{T}_r^{\text{urllc}}$ above 5 Mbps. SOA's throughput falls below the requirement threshold after the 1500 timestamp, and almost 306 violations are observed. Similar to Scenario 1, the latency values in Fig. 7(b) show that AQPS maintains remarkably stable latency values below 0.24-ms throughout most of the simulation. The DRL and RR solutions perform the worst in terms of meeting requirements $\mathcal{L}_r^{\text{urllc}}$ [Fig. 7(e)]. In contrast to Scenario 1, where all solutions have zero packet loss rate for URLLC slice users, DRL, RR, and SOA solutions observe significant $\mathcal{P}_r^{\text{urllc}}$ violations, as shown in Fig. 7(f). The packet loss performance in Fig. 7(c) demonstrates AQPS's excellence, with near-zero packet loss throughout the simulation. The BE performance metrics in Fig. 8(a) and (b) for Scenario 2 reveal significant differences in resource allocation efficiency between the tested solutions. DRL and RR solutions allocate most of the resources to the best-effort traffic and observe no SLA violations for the BE metrics [Fig. 8(c) and (d)] while sacrificing eMBB and URLLC services. In contrast, AQPS intelligently allocates radio resources to BE slice users and experiences 0.0% packet loss rate (0 $\mathcal{P}_r^{\text{urllc}}$ violation) in comparison to 0.85% (249 $\mathcal{P}_r^{\text{urllc}}$ violations) of the SOA solution.

The average performance values of four RRA solutions are shown in Fig. 9. For eMBB traffic in Fig. 9(a), bfAQPS achieves the highest throughput at 62.06 Mbps, surpassing SOA, RR, and DRL by 12.51%, 33.14%, and 45.22%, respectively, while maintaining the lowest latency (0.08 ms) tied with SOA and minimal packet loss (0.08%). Similarly, AQPS delivers 12.24-Mbps throughput for URLLC services [Fig. 9(b)] with a latency value of just 0.09 ms and 0% packet loss, in comparison to DRL, which experiences 0.2% packet loss and SOA with concerning 0.85% loss rates. Fig. 9(c) results demonstrate BE traffic handling performance where AQPS achieves superior resource balancing with 22-Mbps long-term throughput and 29.81-Mbps fifth-percentile throughput, outperforming SOA's fifth-percentile value by 66.33% while showing more consistent performance distribution compared to those of the DRL and RR solutions.

3) *SLA Assurance in Scenarios 1 and 2:* Table III compares the SLA performance metrics for different resource allocation implementations in both scenarios. The proposed AQPS solution achieves superior performance with considerably fewer violation timestamps for all users and experiences 500 violations in Scenario 1 and 1306 in Scenario 2 compared to alternative approaches. This enables exceptional SLA assurance rates of 99.07% and 97.58% for Scenarios 1 and 2, respectively. In contrast, DRL exhibits the lowest performance with SLA assurance rates of only 91.72 % (Scenario 1) and 86.37% (Scenario 2). While both SOA (96.12% and 96.58%) and RR (95.61% and 93.35%) solutions show relatively acceptable performance, they still fail to meet the consistent reliability achieved by the AQPS solution, particularly under the resource-constrained conditions of Scenario 2. This is because the QoS-driven RBG estimation and dynamic urgency-based prioritization in AQPS enable precise resource allocation while maintaining minimal SLA violations. The DRL data-driven mechanism struggles with rapidly adapting to updated requirements under changing network conditions. RR, on the other hand, performs a cyclic static allocation strategy. The performance of the optimized SOA approach indicates that minimum resource computation and allocation in diverse requirements scenarios may lead to suboptimal resource distribution.

4) *Radio Resources Allocations in Scenarios 1 and 2:* The results in Fig. 10 show the dynamic RBG allocation patterns of the four evaluated solutions (AQPS, SOA, RR, and DRL) over the 2000-ms simulation period for scenarios 1 and 2. The resource allocation ($\mathcal{R}$) to different slices reveals different

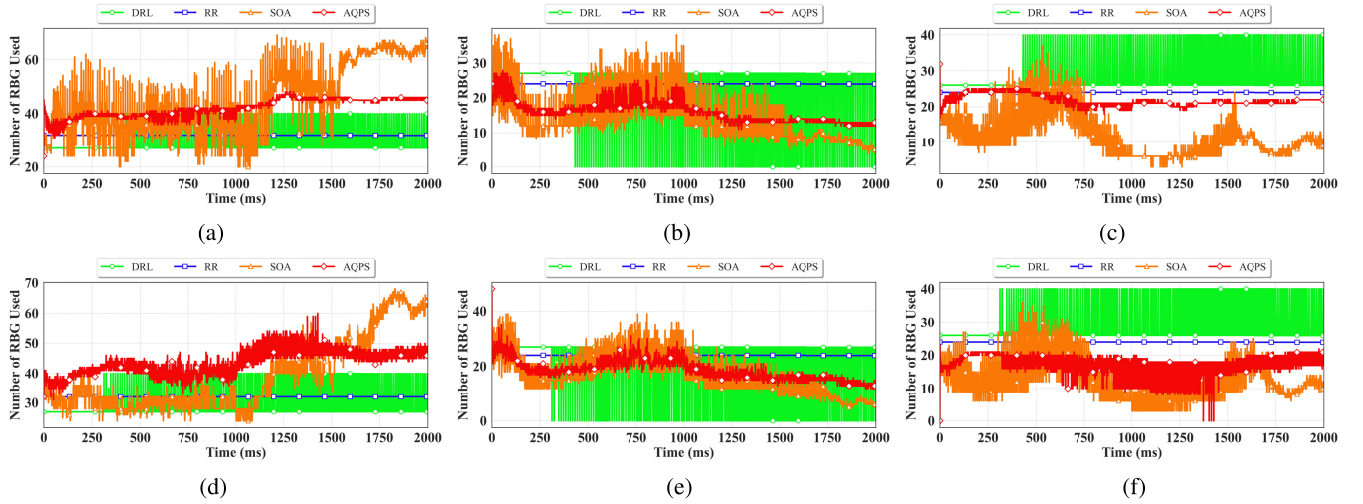


Fig. 10. Per-TTI RBG allocations for eMBB, URLLC, and BE slice users across AQPS, DRL, RR, and SOA resource allocation solutions in Scenarios 1 and 2. (a) S1: $\mathcal{A}_s^{\text{embb}}(t)$ allocations. (b) S1: $\mathcal{A}_s^{\text{urllc}}(t)$ allocations. (c) S1: $\mathcal{A}_s^{\text{be}}(t)$ allocations. (d) S2: $\mathcal{A}_s^{\text{embb}}(t)$ allocations. (e) S2: $\mathcal{A}_s^{\text{urllc}}(t)$ allocations. (f) S2: $\mathcal{A}_s^{\text{be}}(t)$ allocations.

SLA fulfillment strategies employed by each solution. Considering the higher throughput needs of eMBB users, AQPS implements superior resource management by allocating a higher percentage of resources to eMBB services [30.53% in Scenario 1-Fig. 10(a) and 41.55% in Scenario 2-Fig. 10(d)]. At the same time, the proposed AQPS solution maintains sufficient resources for URLLC (11.60% and 17.99%) and BE users (15.84% and 17.38%). This balanced sharing helps AQPS to keep steady QoS across all service types, especially in Scenario 2 [Fig. 10(d)–(f)], where resource demands are much higher. DRL and RR, however, share similar allocation patterns across slices and struggle to prioritize critical services suitably. SOA switches more dynamically with the preference for eMBB services at the expense of URLLC and BE traffic, which leads to severe performance degradation for these services during periods of network congestion.

In summary, RR requires minimal setup implementation with low processing demands and shares resources fairly based on user count. However, it ignores QoS requirements and is unable to adapt to changing network conditions, and therefore, it drops throughput, latency, or packet loss performance. On the other hand, SAC-DRL has the potential to adapt to changing network conditions by finding optimal policies over time. However, the pretrained best-fit model is potentially overfitting in the implemented IoT scenarios. SOA considers QoS requirements and uses a window-based approach to adapt to resource allocations in variable network dynamics. However, in some scenarios, the performance achieved is not optimal. In contrast, the higher performance results of the proposed AQPS show that an SLA-aware priority-oriented scheduling solution adapts well to dynamically changing network conditions and ensures that service requirements are met.

C. Ablation Study

1) *Scenario 3: Validation Using an Alternative Configuration:* We further evaluated the performance of AQPS under an

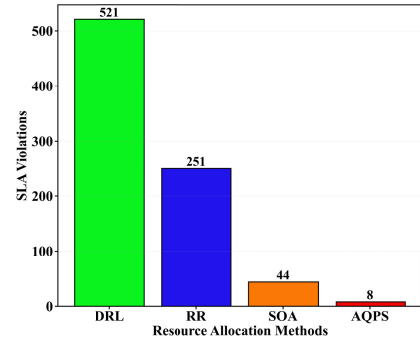


Fig. 11. Total SLA violations for experimental settings defined in [24], and [23].

alternative configuration (Scenario 3) using the slice-specific SLA requirements from [23], [24].

- 1) *eMBB Slice:* $\mathcal{T}_r^{\text{embb}} = 10$ Mbps, $\mathcal{L}_r^{\text{embb}} = 20$ ms, $\mathcal{P}_r^{\text{embb}} = 20\%$.
- 2) *URLLC Slice:* $\mathcal{T}_r^{\text{urllc}} = 1$ Mbps, $\mathcal{L}_r^{\text{urllc}} = 1$ ms, $\mathcal{P}_r^{\text{urllc}} = 0.001\%$.
- 3) *BE Slice:* $\mathcal{G}_r^{\text{be}} = 2$ Mbps, $\mathcal{F}_r^{\text{be}} = 5$ Mbps.

The SLA requirements in Scenario 3 represent a different balance of service demands compared to Scenarios 1 and 2 for eMBB, URLLC, and BE services. Fig. 11 results illustrate the total SLA violations experienced by all RRA approaches for Scenario 3. The violation results demonstrate that AQPS significantly outperforms SOA, RR, and DRL solutions by observing only eight violations for all slice users. In contrast, SOA has 5.5 times, RR has 31.4 times, and DRL has 65.1 times higher violations compared to AQPS. This is because the QoS-driven RBG estimation and dynamic urgency-based prioritization in AQPS effectively adapt to diverse requirement configurations without requiring scenario-specific parameter tuning.

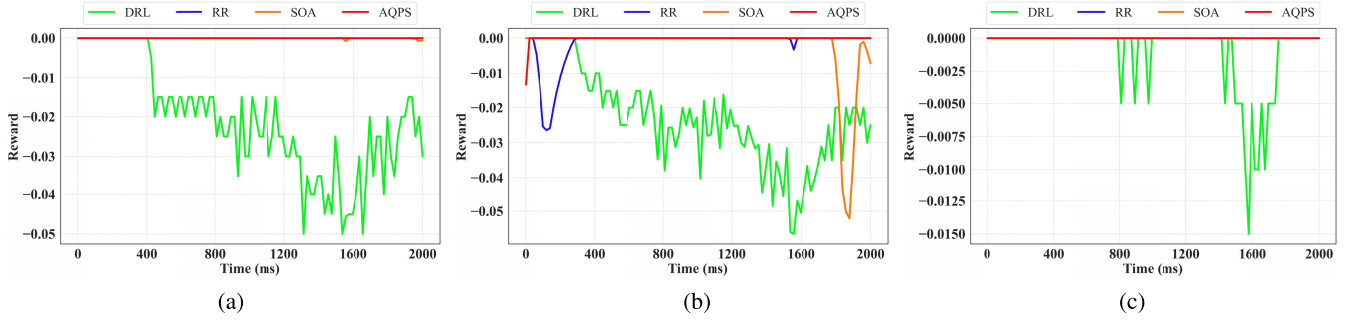


Fig. 12. Per-TTI observed reward values for eMBB, URLLC, and BE slice users across AQPS, DRL, RR, and SOA resource allocation solutions in Scenarios 1, 2, and 3. (a) Scenario 1: Reward values. (b) Scenario 2: Reward values. (c) Scenario 3: Reward values.

2) *Slice-Specific Reward Evaluation*: Next, we evaluated all four RRA solutions using the reward function defined in [24]

$$R(t) = R_{s^{embb}}(t) + R_{s^{urllc}}(t) + R_{s^{be}}(t) \quad (26)$$

where $R_s(t)$ represents the slice-specific reward at time t that captures QoS satisfaction, i.e., throughput, latency, and packet loss for eMBB and URLLC slices, and long-term and fifth-percentile throughput for the BE slice, with a maximum reward value of 0 [23], [24]. Fig. 12 illustrates the reward values over each time slot for Scenarios 1, 2, and 3.

Scenario 1 [Fig. 12(a)] and Scenario 3 [Fig. 12(c)]: AQPS indicates an excellent SLA satisfaction by maintaining an almost-zero reward value throughout the entire 2000-ms simulation period. In contrast, DRL exhibits substantial negative rewards, fluctuating between -0.015 and -0.050 after 400 ms, RR remains stable, and SOA observes some negative values during the third half of the simulation period.

Scenario 2 [Fig. 12(b)]: AQPS shows improved reward performance under the enhanced requirements of Scenario 2. Despite being trained to optimize the same reward function used for evaluation, the DRL method consistently shows negative rewards between 300 and 2000 ms. Notably, SOA experiences significant reward penalties (-0.053) near the end of the simulation (around 1800–2000 ms), while RR shows minor violations early in the simulation. In general, AQPS maintains near-zero reward values for all three scenarios with different SLA configurations.

3) *Performance Impact of Parameters*: Fig. 13 illustrates the performance of AQPS under different values of balance parameter (α_s) and service-specific priority coefficients (ψ_s). The SLA violation counts in Fig. 13(a) shows that the original configuration ($\alpha_{s^{urllc}} = 0.8$, $\alpha_{s^{embb}} = 0.5$, and $\alpha_{s^{be}} = 0.2$) and ($\psi_{s^{urllc}} = 0.545$, $\psi_{s^{embb}} = 0.273$, and $\psi_{s^{be}} = 0.182$) produces only 500 total violations across the 2000 TTI simulation period. Setting the balance parameter for the combined resource share calculation in (23) α_s to 0.5 results in double violations (1012). Similarly, enforcing a uniform service priority coefficient $\psi_s = 0.33$ for all slices results in 1258 violations. This demonstrates that service priority coefficients are the primary mechanism for enforcing SLA differentiation across slice types. Finally, the modification of both parameters (α_s) and (ψ_s) results in 1450 violations.

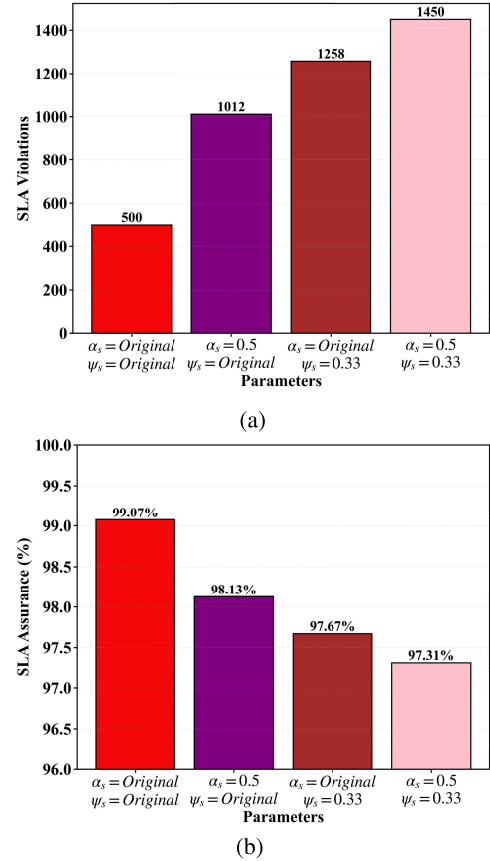


Fig. 13. (a) Total SLA violations and (b) SLA assurance values achieved by AQPS solution when tested under different values of ψ_s and α_s .

Fig. 13(b) illustrates the SLA assurance values for all four combinations of parameter configurations. The achieved SLA assurance values are as follows.

- 1) ($\alpha_{s^{urllc}} = 0.8$, $\alpha_{s^{embb}} = 0.5$, $\alpha_{s^{be}} = 0.2$) and ($\psi_{s^{urllc}} = 0.545$, $\psi_{s^{embb}} = 0.273$, $\psi_{s^{be}} = 0.182$): 99.07%.
- 2) ($\alpha_{s^{urllc}}, \alpha_{s^{embb}}, \alpha_{s^{be}} = 0.5$) and ($\psi_{s^{urllc}} = 0.545$, $\psi_{s^{embb}} = 0.273$, $\psi_{s^{be}} = 0.182$): 98.13%.
- 3) ($\alpha_{s^{urllc}} = 0.8$, $\alpha_{s^{embb}} = 0.5$, $\alpha_{s^{be}} = 0.2$) and ($\psi_{s^{urllc}}, \psi_{s^{embb}}, \psi_{s^{be}} = 0.333$): 97.67%.
- 4) ($\alpha_{s^{urllc}}, \alpha_{s^{embb}}, \alpha_{s^{be}} = 0.5$) and ($\psi_{s^{urllc}}, \psi_{s^{embb}}, \psi_{s^{be}} = 0.333$): 97.31%.

The above findings show that by employing the original parameter values ($\alpha_{\text{grrlc}} = 0.8$, $\alpha_{\text{gsmmb}} = 0.5$, and $\alpha_{\text{gsbe}} = 0.2$) and ($\psi_{\text{grrlc}} = 0.545$, $\psi_{\text{gsmmb}} = 0.273$, and $\psi_{\text{gsbe}} = 0.182$) a well-balanced performance is obtained. In addition, modified parameter values still result in lower numbers of SLA violations and higher SLA assurance compared to SOA (2096, 96.12%), RR (2370, 95.61%), and DRL (4472, 91.72%) solutions.

VI. CONCLUSION AND FUTURE WORKS

In this article, we proposed a novel AQPS solution that addresses efficient RRA problem in a 3GPP-based 5G RAN network slicing environment for IoT applications. In contrast to simpler throughput-based calculations in SOA or complex training and hyperparameter tuning in SAC-DRL methods, the proposed solution is a low-complex multiphase user- and slice-urgency-based resource allocation approach that considers both immediate demands and long-term service quality assurance for the efficient distribution of limited RBGs ($\mathcal{R} = 80$) in IoT-dense urban ($\mathcal{R} = 138$) and enhanced IoT coverage zone ($\mathcal{R} = 104$) scenarios. AQPS ensures efficient resource allocation and avoids SLA violations in highly dynamic resource-constrained wireless environments. Simulation results using three traffic types with variable packet sizes and Poisson flow arrival rates in realistic testing environments with practical spectral efficiency and channel conditions demonstrated the superior performance of AQPS compared to that of alternative algorithms. Specifically, AQPS achieves exceptional SLA assurance rates of 99.07% and 97.58% in both testing environments, with minimum violation timestamps, compared to SOA (96.12% and 96.58%), RR (95.61% and 93.35%), and DRL (91.72% and 86.37%). In the future, AQPS's performance could be improved by incorporating dynamic slice creation and termination in mixed numerology deployments where different slices simultaneously utilize distinct subcarrier spacings, and where numerology-aware resource mapping could better handle heterogeneous resource grid structures.

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